STORAGE



Adopting High-Capacity HDDs: Navigating Throughput and IOPS Limits with SAS4 and NVMe in the AI Cloud



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High-Capacity HDDs - Two IO Limits

Capacity grows faster than IO

HDDs are growing from 30+TB and beyond. IO per TB falls; performance can run out before capacity.

Throughput: the interface ceiling

As media throughput rises, SATA becomes a bottleneck. Faster links are needed to carry the bytes.

Random IOPS: the actuator ceiling

A single actuator limits random IO. More capacity does not add IOPS; a faster interface does not shorten seeks. The ratio of IOPS to capacity will not keep pace as size increase.

Design capacity and IO together

Scale interface bandwidth and device parallelism together. More actuators cannot remove a link bottleneck.

Azure Storage portfolio | Critical for AI

Intelligent data platform underpinning AI workloads—Optimized for AI Training & Inferencing

Powering OpenAI training with Petabits/sec scale

Largest knowledge source for the new Foundry IQ service & Azure AI Search

Supporting AI artifacts at exabyte scale across products like ChatGPT

With the broadest cloud storage portfolio engineered for every AI workload

Block Storage

- Azure Disks
- **Azure Elastic SAN**
- **Azure Container Storage**

Delivers

Persistence for inference state, vector DBs, and training checkpoints

Object Storage

- **Azure Blob**
- **Azure Data Lake**

Delivers

Exabyte-scale training data, model artifacts, and the source behind 95% of vector content

File Storage

- **Azure Files**
- **Azure NetApp Files**
- **Azure Managed Lustre**

Delivers

Tbps-scale throughput keeping GPU clusters fed during training and fine-tuning

Hard Drive - IOPS, Throughput heavy based on workload characteristics. Capacity drives COGS.

OpenAI transforms AI model development with Azure Blob Storage

Challenge

- Checkpoints need exabytes of capacity and high-throughput
- Lost training time costs of $ millions/day

Solution

- Blob Storage - many Exabytes of capacity
- Blob Storage Scaled Accounts - 100s of Tbps throughput

Impact

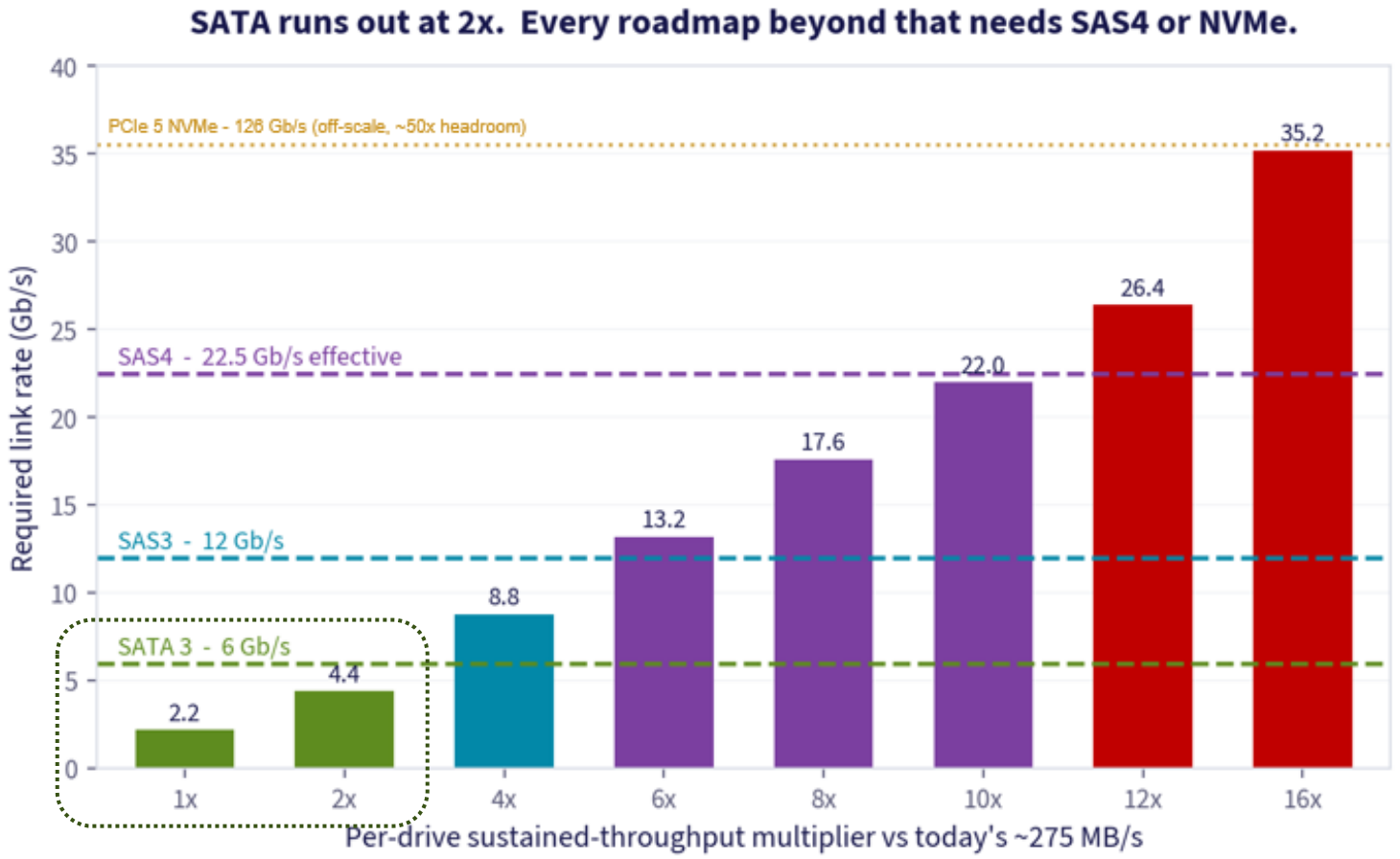
- Increased efficiency with minimal operational overhead
- Durable storage of exabytes of data with high throughput
- Improved data management and disaster recovery



Fairwater supercomputer uses Blob Storage as its filesystem
https://aka.ms/InsideFairwater

Building a DC of this size needs the highest HDD size with supporting IOPS & Throughput

Per Drive: SATA runs out at roughly 2× today's media rate



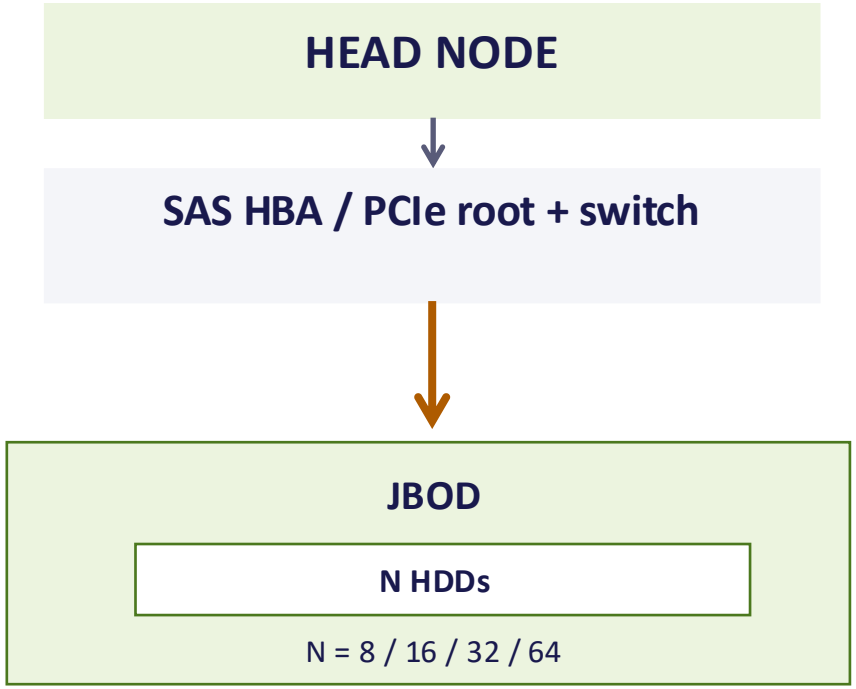
INTERFACE CEILINGS

Interface	Link rate	Multiplier
SATA 3	6 Gb/s	2×
SAS3	12 Gb/s	5×
SAS4	22.5 Gb/s	10×
PCIe 5 NVMe x4	126 Gb/s	~50×
PCIe 6 NVMe x4	252 Gb/s	~100×

System architecture will decide the drive interface

Interconnect for high-capacity drives → SAS or NVME?

HEAD NODE > HBA > JBOD



ORANGE: shared uplink bottleneck

Highest supported throughput multiplier

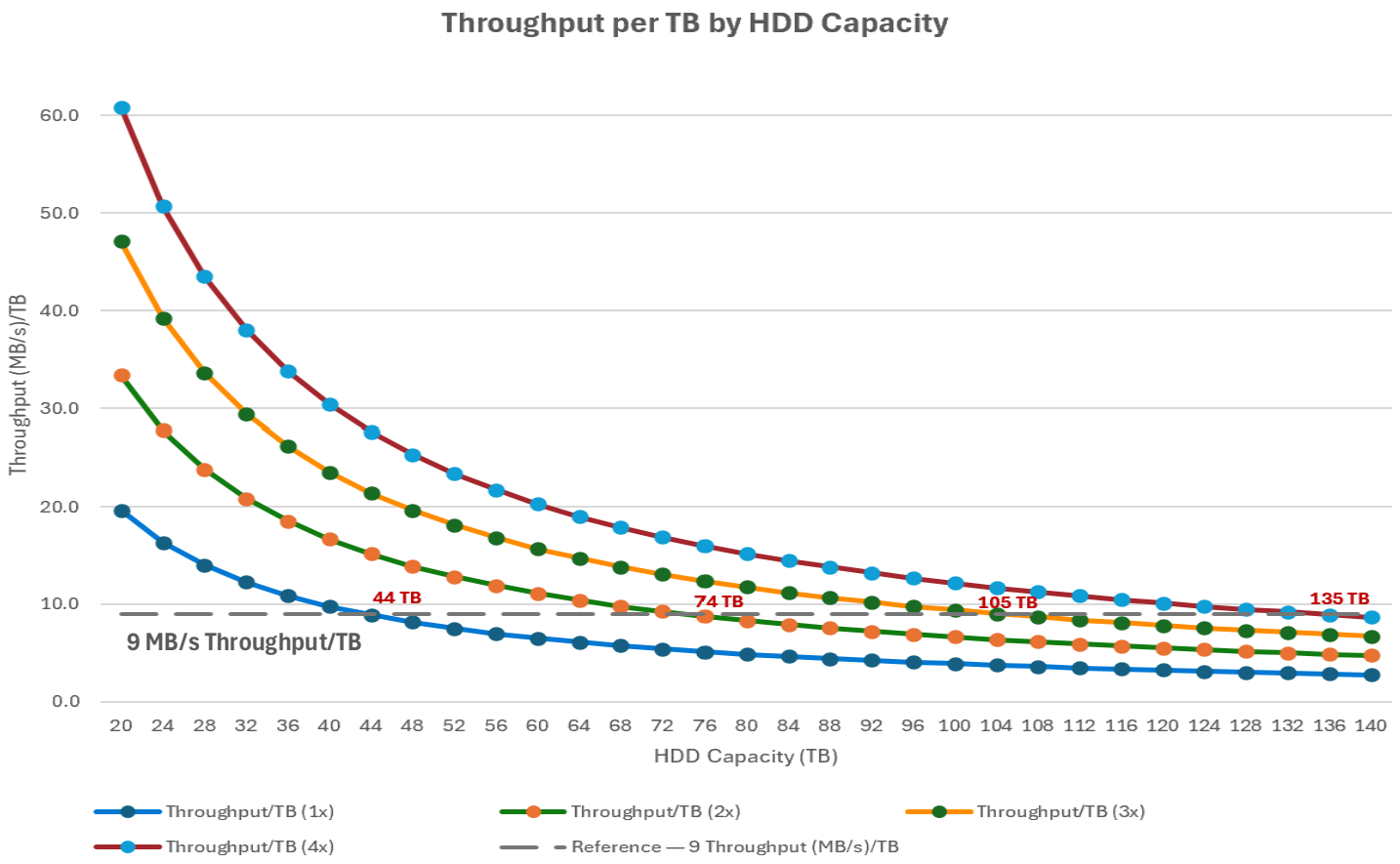
N drives	SAS3	SAS4	PCIe5	PCIe6
8	*4x*	*10x*	*12x*	*16x*
16	*2x*	*4x*	*6x*	*12x*
32	*1x*	*2x*	*3x*	*6x*
64	*<1x*	*1x*	*1x*	*3x*
	SATA/SAS Drive		NVME Drive	

Aggregate-link model | 275 MB/s per drive at 1x multiplier

8 lane connectivity to JBOD

System Architecture Drives → SAS or NVME at higher throughput multipliers

Throughput/TB: Scaling on sequential throughput

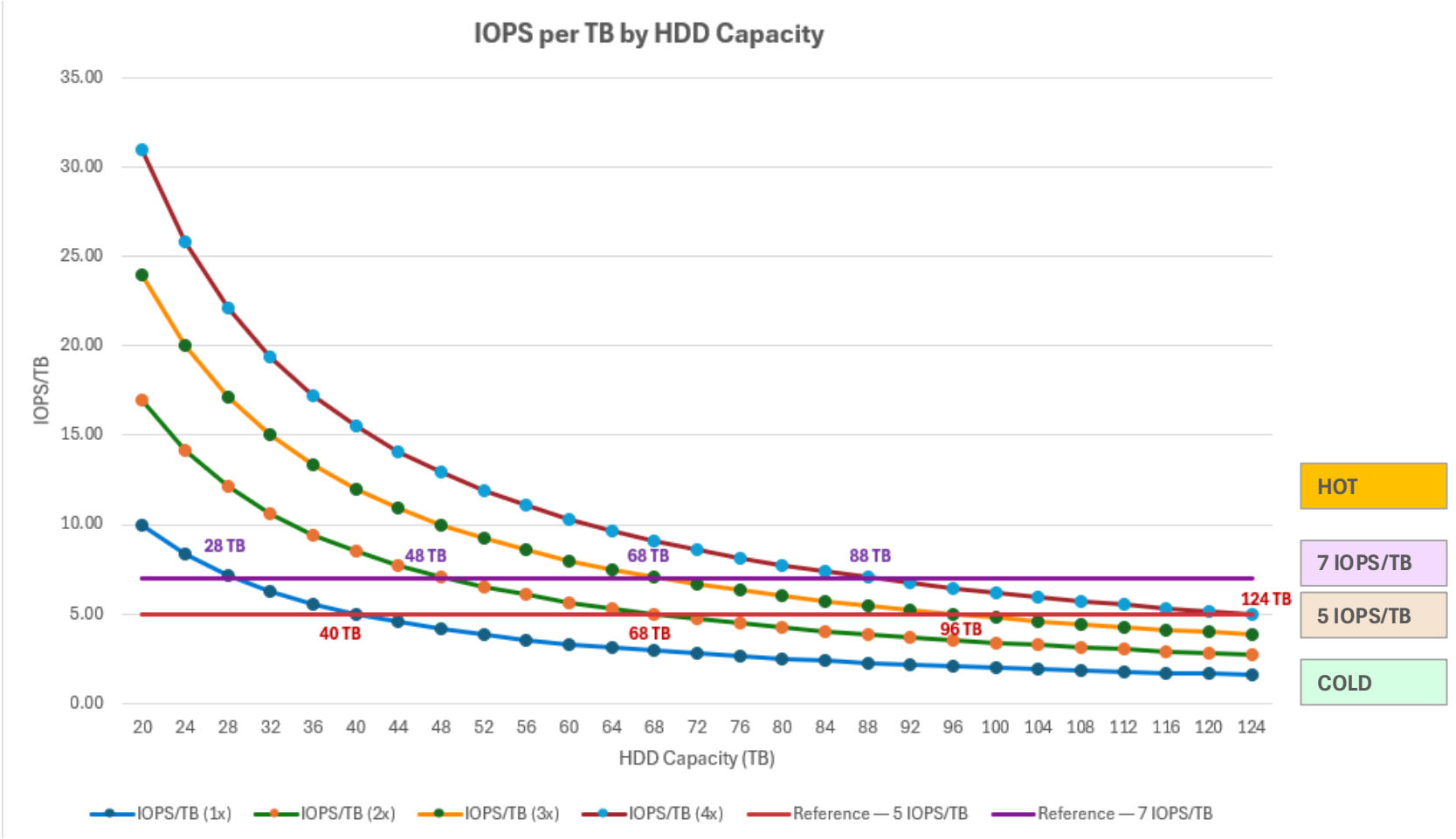


Throughput scaling	9 MBps/TB
1x	44
2x	74
3x	105
4x	135

Target 9 MBps throughput/TB as capacity scales

Assumes 275 MBps baseline, 70% HDD fill , 70% efficiency with incremental throughput multiplier

IOPS/TB: Scaling IOPS



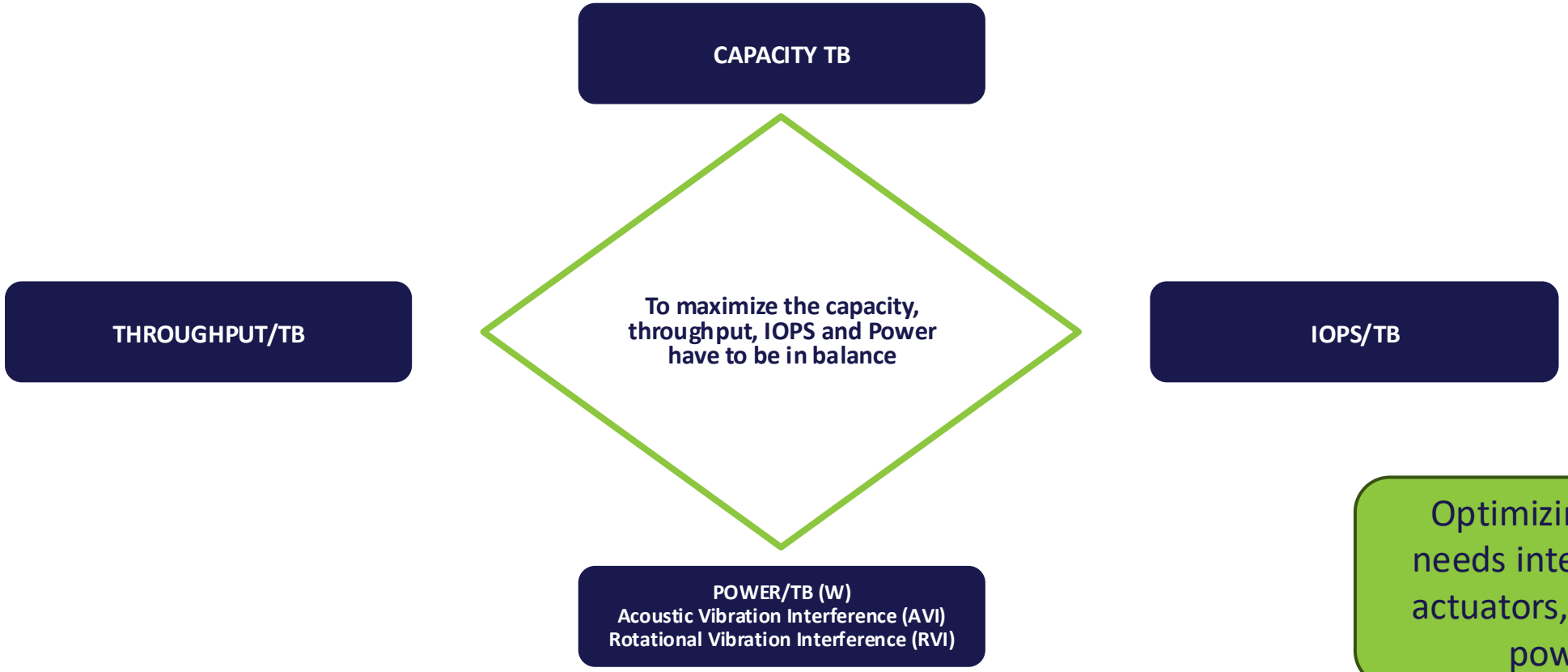
IOPS scaling	5 IOPS/TB HDD TB	7 IOPS/TB HDD TB
1x	40	28
2x	68	48
3x	96	68
4x	124	88

Target 5 - 7 IOPS/TB as capacity scales

Assumes 140 IOPS baseline, 70% HDD fill , 70% efficiency with incremental IOPS multiplier

The four-way trade-off

Capacity, throughput, IOPS, and power all pull against each other as capacity scales.



Optimizing all 4 vectors needs interface adoption, actuators, performance & power tuning

Supply Chain and Validation Impact

New silicon cycles

Controller redesigns and physical interface changes ripple into manufacturing and test infrastructure.

Multi-vendor compatibility

Components from different vendors must interoperate cleanly or adoption stalls at integration.

Coordination beats going it alone

Manufacturers, integrators, and standards bodies need a shared plan to avoid duplicated effort.



Call to Action And OCP Collaboration Opportunities

Standardize HDD over SAS4 and NVME

Define how high-capacity drives expose and negotiate throughput capability across the interfaces. Future proof with NVME.

Expose IOPS multiplier capability

Agree on a discovery and capability model so hosts can schedule against real device parallelism.

Share validation frameworks

Common interoperability tests reduce duplicated qualification work across the ecosystem.

Publish benchmarks and telemetry

Real-world workload traces and open IO telemetry let everyone design against the same evidence.

Let us make higher HDD capacities usable, interoperable, and economical across the ecosystem.

Thank You

